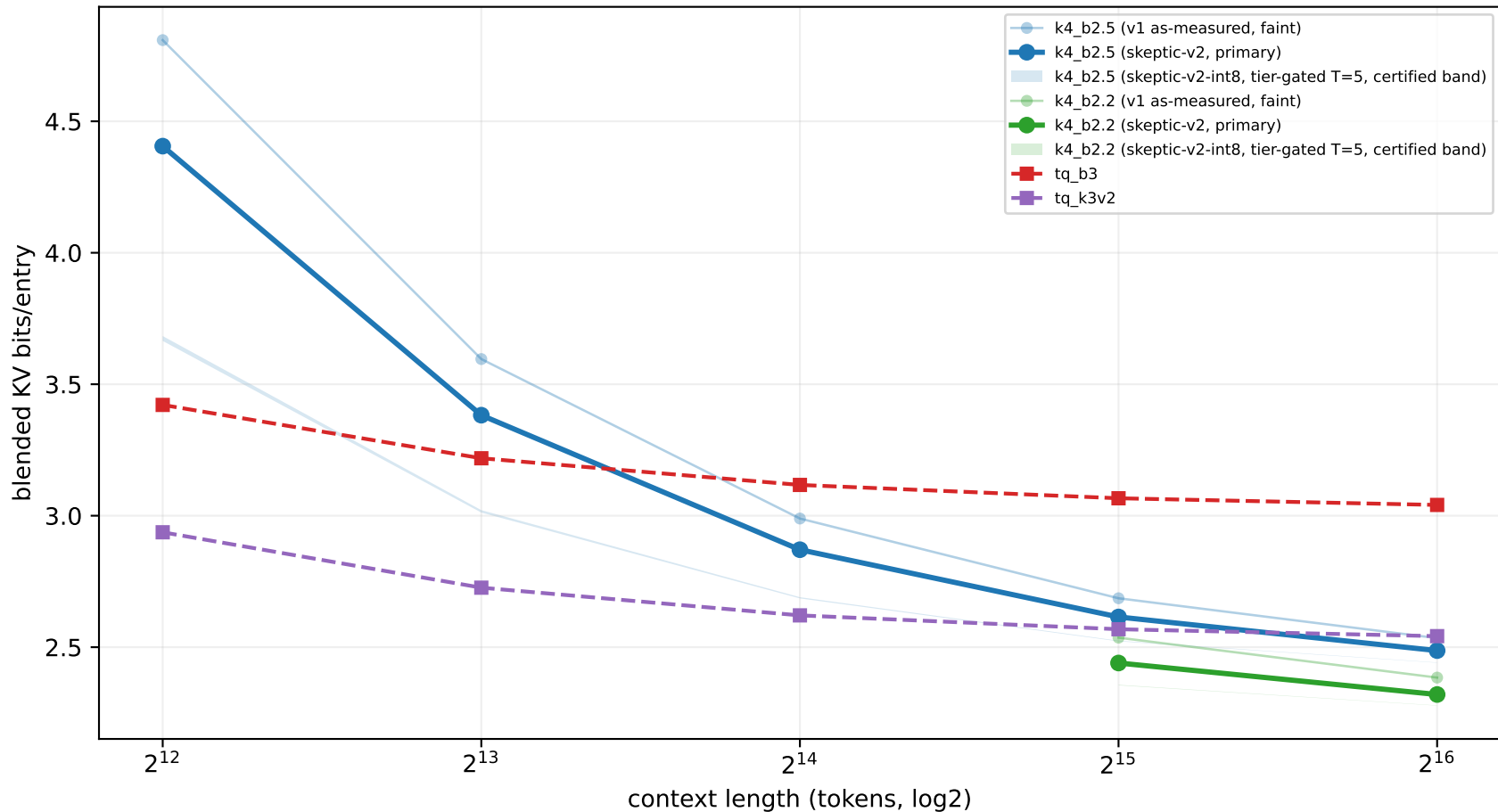


K4 bits-vs-context (Llama-3.1-8B-Instruct, real-text NIAH duel)



Model: meta-llama/Llama-3.1-8B-Instruct. Solid = skeptical-v2 (primary accounting, used-decoder-columns only); faint = skeptical-v1 as-measured (full-C fp16 decoder charge, every parquet before 2026-07-23). The blanket skeptical-v2-int8 projection this figure used to draw as a dashed line was REJECTED by its own certificate + measurement (rel_degradation 13.5%/16.7% at b2.2/b2.5 vs the 5% line, docs/2026-07-25-k4-local-levers-results.md); the shaded band replaces it with the CERTIFIED tier-gated (T=5) accounting -- mixed_dec_charge at int8 charge-saving fraction 0.893/0.916 (the gpt2 T=5 sweep's per-budget fracs, applied as a band to each arm), measured rel_degradation 0.50%/0.60% at b2.2/b2.5, well inside the 5% gate. Band is a gpt2-fit ESTIMATE pending exact Llama tier counts at the refit. TQ baseline rows carry no spectral pack and are accounting-mode-invariant. k4_b2.5 x tq_b3 crossover -- v1: between 8192 and 16384 (~13,076); v2: between 8192 and 16384 (~10,236) | k4_b2.5 x tq_k3v2 crossover -- v1: between 32768 and 65536 (~61,820); v2: between 32768 and 65536 (~42,522). docs/2026-07-15-k4-duel-results.md, docs/2026-07-25-k4-local-levers-results.md.